

Project Design Phase

Solution Architecture

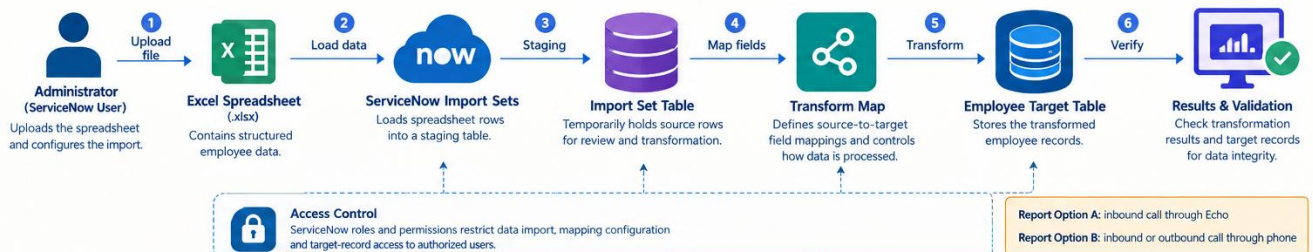
Date	01 October 2026
Team ID	SWTID-2026-9203
Project Name	Import Data using Transform Maps (Spreadsheet)
Maximum Marks	4 Marks

Solution architecture bridges the business problem and the technology solution. The supplied template expects the architecture to describe the structure, characteristics, behaviour, features and specifications of the proposed solution.

Architecture Diagram

Solution Architecture and Data Flow – Importing Employee Spreadsheet Data into ServiceNow

The architecture bridges the business problem and the technology solution by showing how employee data flows from a spreadsheet to the ServiceNow target table using Import Sets and Transform Maps.



Architecture Components		
Component	Technology / Service	Role in the Solution
External Data Source	Excel Spreadsheet (.xlsx)	Contains structured employee data received for bulk migration.
Data Loading	ServiceNow Import Sets	Loads spreadsheet rows into a staging/import table before they affect target records.
Staging Layer	Import Set Table	Temporarily holds source rows so the data can be reviewed and transformed.
Transformation Layer	ServiceNow Transform Map	Defines source-to-target field mappings and controls how staged data reaches the target table.
Target Data	ServiceNow Employee Target Table	Stores the transformed employee records.
Verification	Transform Results / Target Record Review	Confirms successful processing and allows incorrect or failed records to be identified.
Access Control	ServiceNow roles and permissions	Restricts data import, mapping configuration and target-record access to authorized users.

Source alignment: This solution architecture follows the same principle as the provided template – showing the bridge between the business problem and the technology solution using ServiceNow-specific components.

End-to-End Data Flow	
1	Employee data is prepared in an external spreadsheet.
2	The spreadsheet is loaded into a ServiceNow Import Set.
3	The Import Set table acts as the staging area for source rows.
4	A Transform Map maps source fields to corresponding target-table fields.
5	The transformation is executed to create or update employee target records.
6	Transformation results and target records are reviewed to validate data integrity.

Architecture Characteristics	
Characteristic	Project-specific Description
Reliability	Staging and explicit field mapping provide a controlled path between source data and target records.
Usability	The workflow is suitable for beginners because it separates loading, mapping, transformation and verification into clear steps.
Security	Import and target-record operations should be restricted to authorized ServiceNow users.
Scalability	The same pattern can be reused for larger batches and other structured data sets by changing mappings and target configuration.

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Source alignment: The supplied Solution Architecture template presents architecture as the bridge between business problems and technology solutions and illustrates architecture/data flow with a diagram. This project follows the same principle using ServiceNow-specific components.